SuperMimic on Kangaroo, M4: the stability study

Why the online loop oscillated, five 12 h runs on Leonardo to find out, and what we changed. Branch `omm/supermimic` of pal_mjlab, 26 October 2026.

Summary

- **The oscillation came from an early critic blow-up, and two small changes prevent it.** In the first 2000 iterations a run either blew up its critic (value down to -175 to -396, loss up to 10^8) or did not. The two runs whose policy uses that critic and blew it up (baseline, A6) kept spiking afterwards. No run that avoided the blow-up ever spiked. A1 blew up its critic too, but its policy does not use it, and it stayed stable.
- **B, A2′ and A5 trained without a single collapse** in 12 h: no failure spike or critic-loss burst at any iteration, and no collapse at any of their checkpoints (every 1000 iterations, all benchmarked). Their benchmark rises steadily to tracking return 0.89 with 2–4 failed clips (of 1296). That beats the old baseline's best checkpoint in its full 24 h run (0.872), and it gets there in about 20,000 iterations. The baseline collapsed 3 times over the same iterations.
- **B is the plain algorithm with another seed**: it avoided the blow-up by chance. A2′ (failure head on the rule's two errors) and A5 (critic loss weighted by survival) avoid it by design. A2′ is also the fastest: 4 failures after 2000 iterations, against 48–49.
- **A1 (no critic) is stable but worse**: 16–20 failures at its best and last checkpoints, against 2–4. Without the blow-up the critic is not what makes training unstable, and it helps tracking.
- **A6 (no privileged critic inputs) does not help**: it blew up early and spiked like the baseline (same seed), and reached only 0.711 by iteration 20,000. Stopped at 6 h.
- **Adopted: A2′ and A5 together** as the default (D39), now in the 24 h `full` and `dev` runs. Both are stable at about iteration 18,000 so far.
- One run per configuration. The changes are large and systematic, but the exact numbers of a single run carry seed noise.

1 The question

SuperMimic's online loop oscillated on `dev`, locally (bf16, replay 500) and on the cluster with the paper's settings (TF32, replay 1000; `cluster_report.pdf`). The policy repeatedly collapsed from a few failures to most clips failing, and recovered. Before the runs, a diagnostic on the old checkpoints (`scripts/supermimic/reward_extremes.py`) separated two mechanisms:

1. **The main collapses: the critic goes wrong on live states.** Between the checkpoint before a collapse and the collapse checkpoint, the critic's error on imagined steps before any predicted fall grows 40–150×. The critic overestimates where the robot is still up, and the policy follows its gradient (about 70% of the policy's objective is the critic's value at step 32).
2. **A late burst: post-fall extrapolation.** After an imagined fall the world model is outside its data and its predicted joints run away. The joint-limit penalty, linear and unbounded, reaches -10^6 per step. The policy's objective weights those steps by the survival $\hat{c}_t \approx 0$, but the paper's critic loss (Eq. 3) does not, so a few such rollouts dominate it (loss up to 10^{14}).

2 The runs

One change per run against the cluster baseline (seed 42), all from one commit, 12 h on one A100 each, with the paper's settings and compiled imagination (D37). Every checkpoint (every 1000 iterations) is benchmarked on the same 1296 `dev` clips as all earlier runs, every clip once from its first frame.

run	change	what it tests
B	none, seed 43	run-to-run variance
A1	no critic ($H = 32$)	do the collapses need the critic? (also the paper's Fig. 4b cell)
A2′	failure head on the rule's two errors (root height, gravity-z; 64×64) instead of 256×256 on the full state, as SuperMimic-Reproduced	does the policy exploit the learned head?
A5	critic loss weighted by $\hat{c}_t$ (a deviation from Eq. 3)	does post-fall data poison the critic?
A6	critic without our privileged inputs (root-pose error, time left)	do our critic additions (D7, D22) mislead it? Stopped at 6 h.

3 Training signals

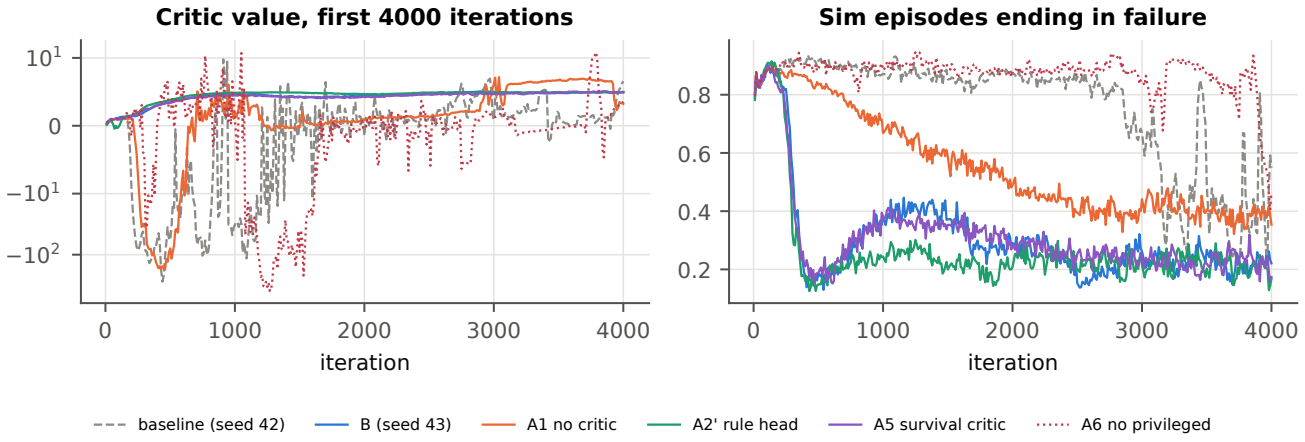


Figure 1: The first 4000 iterations. Left: the critic's value of the imagined start states (symmetric log scale). Right: the share of simulator episodes ending in a failure.

Early on, the runs split in two (Figure 1).

- **A critic blow-up in the first 2000 iterations** hit the baseline, A6 and A1. Their critic's value fell to −277, −396 and −175, and its loss reached 10^6–10^8. These runs took off late: the share of failing episodes first dropped below 50% at iterations 3160, 3980 and 1970. A1's critic is trained but not used by its policy, which is why it recovered sooner.
- **B, A2′ and A5 never had one.** Their value stayed between 0 and 5, and they took off at iterations 310, 290 and 300.
- **Likely mechanism.** At the start, the 256×256 learned head ranks failures worse than chance (AUC about 0.4). So imagined falls keep their survival weight, and the exploding post-fall penalties flood both the policy's objective and the critic's targets.
  - A2′'s two-feature head learns failure almost at once, so falls are cut off.
  - A5 keeps post-fall steps out of the critic's fit.
  - B avoided it by chance; its seed alone made the difference.

Afterwards (Figure 2, up to iteration 35,520, the shortest run):

- **B, A1, A2′ and A5 had no failure spike and no critic-loss burst at all.** Their critic loss stayed near 10^{-2} and their policy gradient near 0.5 throughout. Imagination tracked the simulator at a correlation of 0.99 or more at every check.
- **The baseline** had 7.2 failure spikes per 10,000 iterations and 15 critic-loss bursts over the same range. **A6** had 5.8 spikes per 10,000 before it was stopped. Both had checks where imagination and simulator disagreed (correlation down to 0.15 and 0.66).

- **A1 is calm but tracks a little worse.** Over the last 10,000 iterations, 29% of its training episodes end in a failure, against 20–21% for the others; its 32-step failure rate in the simulator is 3.5%, against 1.1–1.6%. Without the critic the policy is more myopic.
- **Extreme imagined rewards still occur** in every variant, because the post-fall extrapolation is untouched (iterations with a reward below −1: A5 14, B 25, A2′ 169, A1 676, A6 560). In the stable runs they no longer reach the critic in force: A5 down-weights them, and A2′'s sharp head stops them from propagating backward through the λ-returns.

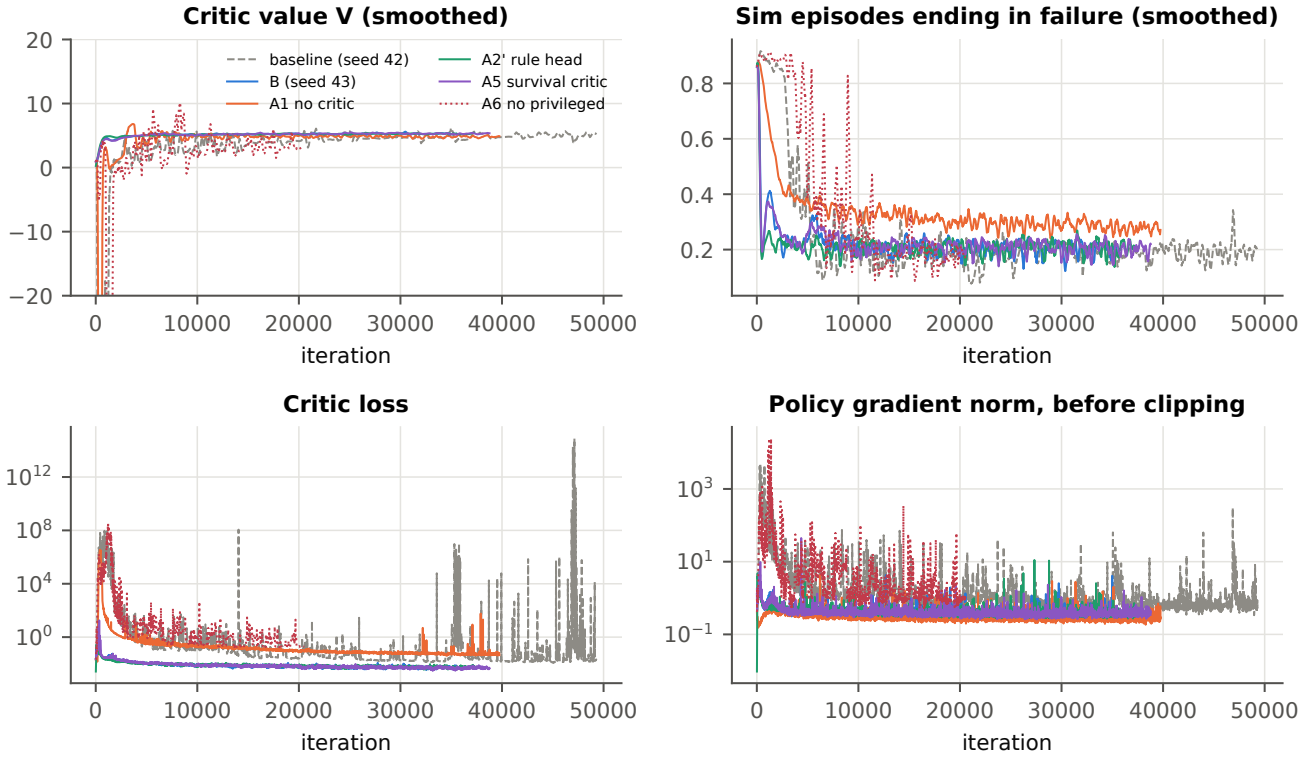


Figure 2: All iterations. Critic value and the share of failing episodes are moving averages; the critic loss and the policy gradient norm (before clipping) are raw.

4 Benchmark

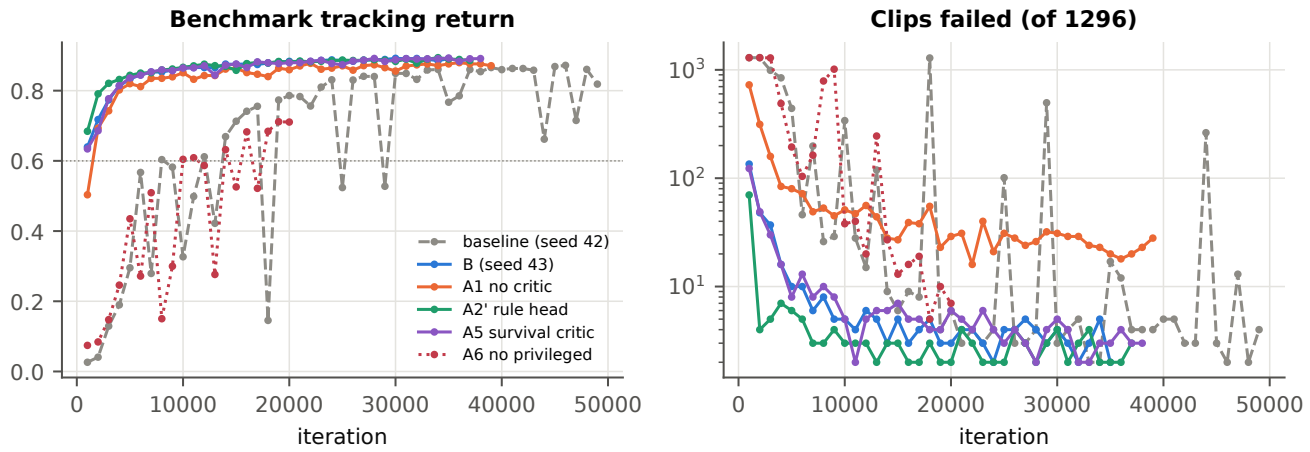


Figure 3: The benchmark of every checkpoint (1296 `dev` clips, each once from its first frame). The failure axis is logarithmic; zero failures are drawn at 1. The baseline is shown over its whole 24 h run; the table cuts every run at iteration 35,520.

Table 1: Per run, up to the shortest run's last iteration (A6: until it was stopped). Early min V: the critic's lowest value in the first 3000 iterations. Take-off: the first checkpoint at PPO-Sim's local 12 h tracking return (0.742). Best and last: failures / tracking return. Collapses: checkpoints with tracking return below 0.6 after take-off. Spikes: onsets of training-episode failure spikes after iteration 5000. Critic bursts: logged critic losses above 10^3.

run	up to	early min V	take-off	best	last	collapses	spikes/10k	critic bursts
baseline (seed 42)	35,520	-277.0	17,000	2 / 0.861	17 / 0.767	3 of 19	7.2	15
B (seed 43)	35,520	0.3	3,000	4 / 0.892	2 / 0.891	0 of 33	0.0	0
A1 no critic	35,520	-175.3	3,000	16 / 0.878	20 / 0.876	0 of 33	0.0	0
A2' rule head	35,520	-0.4	2,000	2 / 0.894	2 / 0.890	0 of 34	0.0	0
A5 survival critic	35,520	0.3	3,000	3 / 0.893	3 / 0.893	0 of 33	0.0	0
A6 no privileged	20,530	-395.8	–	10 / 0.711	7 / 0.711	0 of 0	5.8	1

- **No collapse in B, A2′, A5 or A1**, at any of their 33–34 checkpoints after take-off. Between consecutive checkpoints, the tracking return never drops by more than 0.026. From iteration 10,000 on, no checkpoint of B, A2′ or A5 fails more than 6, 4 and 7 clips, or scores below 0.844. Over the same 35,520 iterations, the baseline collapsed 3 times: 1287 failures at 18,000, and checkpoints below 0.6 at 25,000 and 29,000.
- **Fast take-off.** B, A2′ and A5 pass PPO-Sim's local 12 h level (0.742) by iteration 3000, A2′ by 2000. The baseline needed 17,000.
- **Better at the end.** B, A2′ and A5 reach 0.890–0.894 with 2–4 failures. The baseline's best, over its whole 24 h run, was 0.872 with 2 failures (iteration 46,000).
- **A2′ leads early**: 4 failures at iteration 2000 and 3 by 8000. B and A5 get down to about 5 failures only by iteration 10,000.
- **A1** improves steadily but stays at 16–159 failures after take-off (16–84 from 4000; tracking return up to 0.878). Without the critic's value beyond 32 steps, the policy is more myopic.
- **A6** never reaches 0.742 before it was stopped: 7 failures, tracking return 0.711 at 20,000.

Two of the baseline's collapses (25,000 and 29,000) showed only at odd checkpoints, which is why every checkpoint is benchmarked here, not only every second one. The training signals, which cover every iteration, agree: no failure spike and no critic-loss burst in these runs.

5 What this means

1. **The instability was a consequence of the early blow-up, not a property of the algorithm.** The same code, with nothing changed (B), trained stably when it happened to avoid the blow-up. The runs that went through one (baseline, A6) kept spiking for tens of thousands of iterations. The damage persists, for example in the critic's weights or in Adam's moment estimates; we did not isolate which.
2. **The cause of the blow-up is the post-fall extrapolation, unweighted.** Early, the learned head misses falls, so exploding post-fall penalties enter the policy's objective and every critic target. A2′ removes the first route by making the survival weights right from the start. A5 removes the second by keeping post-fall steps out of the critic's fit.
3. **Removing our privileged critic inputs does not help.** A6 (root-pose error and time left removed) blew up and spiked like the baseline, which has the same seed. Whether D7 and D22 help on their own is not settled; they stay.
4. **The critic helps tracking.** Without it (A1) training is stable, but the policy fails several times more clips (16–20 against 2–4 at the best and last checkpoints).

6 Also learned along the way

- **The code is not what made the new runs calm.** One update, computed by the old and the new code on identical inputs, agrees to 10^{-6} in FP32. In TF32, rounding alone, amplified through the 32-step backward pass, moves the actor gradient by 3–12%. That changes which chaotic trajectory a run follows, as a seed does, but not its expected behavior.
- **Speed.** Compiling the imagination step and the world-model loss (D37) brings an A100 iteration from 1.78 s to about 1.1 s. CUDA graphs on top (D41) measured 2.8× on the policy update in isolation, but hung in the training loop. They are not used.
- **The return-based sampler concentrates.** Once the policy is good, about half of the starts go to about 1% of the bins. One clip (`jog_butt_kick_R_loop_002`) gets 7–42% of all starts. This is not a collapse trigger, but it heavily skews the training data.
- **Two infeasible clips** sink 6–23 cm below the ground and fail in every good policy. They were removed from the dataset (D40); this report's benchmarks still use the 1296-clip `dev`.
- **The local GPU queue preempted itself.** The queue daemon's game check matched the `pgrep` of concurrent `gpuq` status calls, so polling the queue killed running jobs. Fixed in gpuq: no automatic game detection, the user takes the GPU with `gpuq pause`.

7 Decisions

- **D39, adopted:** the failure head on the rule's two errors (64×64) and the survival-weighted critic loss are the default. A2′ and A5 each prevent the blow-up alone; together they cover both routes. The paper's variants stay available by flag.
- **D40:** the two infeasible clips are out of the dataset.
- **D37:** compiled imagination, about 1.1 s per iteration on an A100. **D41:** CUDA graphs not used until they work in the training loop.
- **Running now** (`docs/supermimic/m5/runs.md`): `full` (124,077 clips) and `dev` (1294 clips), 24 h each with these defaults.
- **Open:**
  - benchmark the 24 h runs;
  - a second seed of the combined default, to put a number on run-to-run noise;
  - the sampler's concentration on a few clips;
  - CUDA graphs in the training loop.

A Reproduction

```
# runs (Leonardo, lazyhpc run folders 20261025-153632_<job>_5eb28ab), one flag each, e.g.
python -m pal_mjlab.scripts.supermimic.train Mjlab-SuperMimic-Flat-Pal-Kangaroo-Grippers-ArmWaist1Hz \
  --env.commands.motion.dataset <dataset> --env.commands.motion.subset dev \
  --agent.experiment-name supermimic_stability --agent.run-name A5-survival \
  --agent.algorithm.critic.survival-weighted True
# diagnostic of old checkpoints; benchmark on the 1296-clip dev
python -m pal_mjlab.scripts.supermimic.reward_extremes --task <task> --checkpoints <run>/model_N.pt --out-dir <dir>
python -m pal_mjlab.scripts.supermimic.eval --task <task> --library <dataset>/libraries_archive/dev_1296 \
  --checkpoints <run>/model_*.pt --out-dir logs/supermimic_eval/stability_<run>
# figures, tables, this report
python docs/supermimic/m4/make_stability_figures.py && (cd docs/supermimic/m4 && latexmk -pdf stability_report.tex)
```